

Architecture Design Report

This document presents the architectural design of the system, including its context, logical structure, runtime behavior, and deployment.

1. System Requirements

System Name: LibraryManagementSystem

Architecture Style: Microservices Architecture

Functional Requirements

- The system shall detect the user's face using the YOLOv5 model.
- The system shall recognize gestures.
- The system shall track the user's hand.
- The system shall support single-hand interaction.
- The system shall allow users to customize actions for specific gestures.
- The system shall enable real-time collaboration by supporting interactions on the projected surface.
- The system shall have low-latency.

Non-Functional Requirements

- The system shall be cost-effective.

Description: The system shall be affordable to many and have no hardware limitations.

- The system shall be flexible.

Description: The system shall work effectively in different places.

- The system shall have a competitive price.

Description: The system shall cost way less than all other systems in the market.

- The system shall have a user-friendly interface.

Description: The system shall be easy to use and have more interactive features than traditional touch projection systems.

- The system shall be secure.

Description: The system shall store user account information in a secure database.

2. Formal Architecture Specification (ACME ADL)

The following section provides a formal architectural specification using ACME ADL, capturing components, connectors, and constraints.

```

System LibraryManagementSystem {
  Property view : String = "runtime";
  Property architectureStyle : String = "Microservices Architecture";
  Property source : String = "AI";
  Component FaceDetectionModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Detect the user's face using the YOLOv5 model, contours detection";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component GestureRecognitionModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Recognize gestures using threading for concurrent execution";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component HandTrackingModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Track the user's hand using Background Subtraction (BS).";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component InteractionModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Support single-hand interaction, exclude multiple-hand interaction";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component CollaborationModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Enable real-time collaboration by supporting interactions on the same device";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component LatencyOptimizationModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Analyze gestures and reflect the corresponding action on the device";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Attachment FaceDetectionModule.out to HandTrackingModule.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
  }
  Attachment HandTrackingModule.out to GestureRecognitionModule.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
  }
  Attachment GestureRecognitionModule.out to InteractionModule.in {
    Property delivery : String = "at-least-once";
  }
}

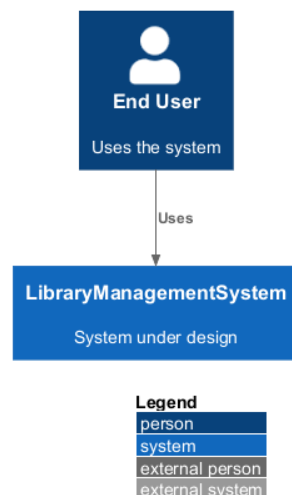
```

```

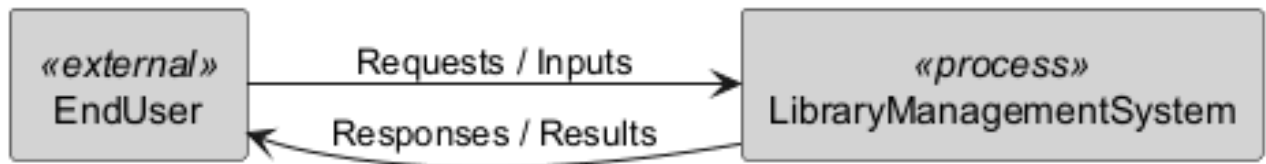
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment InteractionModule.out to CollaborationModule.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment CollaborationModule.out to LatencyOptimizationModule.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment LatencyOptimizationModule.out to InteractionModule.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment FaceDetectionModule.out to GestureRecognitionModule.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Property resilience_pattern : String = "Retry, CircuitBreaker";
Property delivery : String = "at-least-once";
Property ordering : Boolean = false;
Property authentication : String = "service-to-service";
Property authorization : String = "RBAC";
}

```

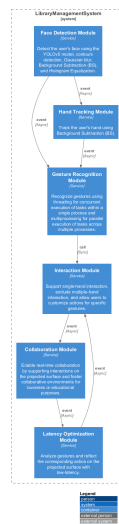
3. Context View



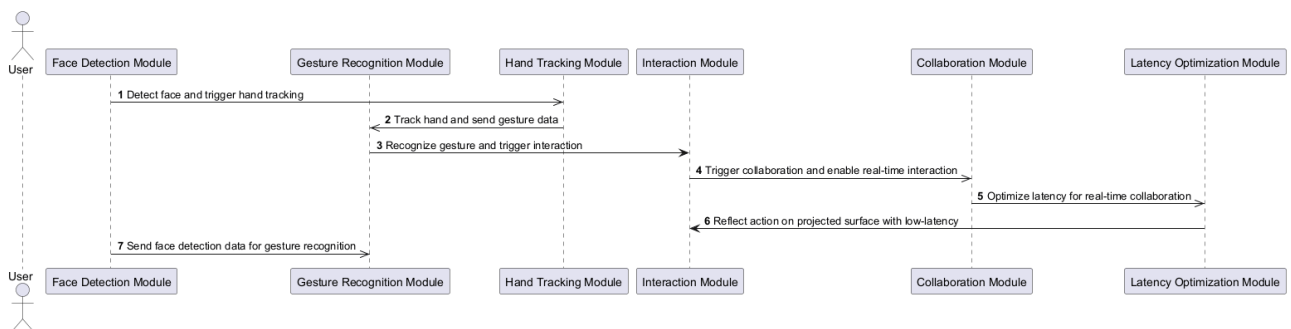
3.1 Data Context View (Level 0 DFD)



4. Logical View (C4 Container Diagram)



5. Process View (Runtime Interaction)



6. Physical View (Deployment Diagram)

